

JEE MAIN REVISION MASTER: CURRENT ELECTRICITY

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 66 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Ohm's Law, Drift Velocity & Resistance Temperature	48 PYQs	Conduction
Very High	Kirchhoff's Laws, Wheatstone Bridge & Potentiometer	20 PYQs	DC Instruments

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Ohm's Law, Drift Velocity & Resistance Temperature	$I = n e A v_d$, $v_d = \frac{e E \tau}{m}$, $\rho = \frac{m}{n e^2 \tau}$, $R(T) = R_0(1 + \alpha \Delta T)$, $\vec{J} = \sigma \vec{E}$
Kirchhoff's Laws, Wheatstone Bridge & Potentiometer	$\frac{P}{Q} = \frac{R}{S}$ (Wheatstone), $\frac{E_1}{E_2} = \frac{I_1}{I_2}$, $r = R \left(\frac{I_1}{I_2} - 1 \right)$, $S = \frac{I_g}{I - I_g}$, $R_{\text{series}} = \frac{V}{I_g} - G$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Ohm's Law, Drift Velocity & Resistance Temperature (Very High Yield — 48 PYQs)

Core Principle: Microscopic model of electrical conduction, electron mobility $\mu = v_d/E$, and thermal variation of resistivity.

■ **Caution / JEE Pitfall:** For conductors, $\alpha > 0$ (resistance increases with temperature). For semiconductors, $\alpha < 0$ (resistance decreases with temperature).

■ **Speed Trick:** When a wire is stretched to n times its initial length while preserving volume, its resistance becomes $n^2 R$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Given below are two statements: Statement-I: The resistivity of a conductor is independent of its temperature. Statement-II: The resistivity..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "The number of electrons moving per second through the filament of a lamp of 60 W operating at 120 V is nearly $(e=1.6 \times 10^{-19})$..." [Ans: C]
- [2026 Shift] Q#3 (MCQ): "In a conducting region, 10^{19} electrons and 10^{19} protons move to the left, while $10^{19} \alpha$ -particles move to the right per..." [Ans: B]

3.2 Kirchhoff's Laws, Wheatstone Bridge & Potentiometer (Very High Yield — 20 PYQs)

Core Principle: Kirchhoff's junction and loop rules, balanced bridges, potentiometer EMF comparison, and galvanometer conversion.

■ **Caution / JEE Pitfall:** A potentiometer measures EMF with zero drawn current, making it infinitely superior to a real voltmeter which always draws finite current.

■ **Speed Trick:** To convert galvanometer to ammeter: connect small shunt S in parallel. To voltmeter: connect high resistance R in series.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "In the figure, the values of currents I_1, I_2 and I_3 respectively are..." [Ans: N/A]
- **[2026 Shift] Q#2 (MCQ):** "Which of the following circuits is correct for verification of Ohm's law?..." [Ans: N/A]
- **[2025 Shift] Q#2 (MCQ):** "In an experiment to determine the figure of merit of a galvanometer by half deflection method, a student constructed the following circuit. ..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

In the figure, the values of currents I_1, I_2 and I_3 respectively are

- (A) 6 A, 1.5 A and 1 A
- (B) 4 A, 2.5 A and 2 A
- (C) 4 A, 2.5 A and 1 A
- (D) 6 A, 4.5 A and 1.5 A

Correct Answer: None

Solution Analysis: At any junction, Kirchhoff's Current Law says that the total current entering the junction is equal to the total current leaving it. For the first junction, 5A enters. The currents leaving are 1A and I_1 . So, $5 = 1 + I_1 \Rightarrow I_1 = 5 - 1 = 4A$ For the second junction, I_1 enters. The currents leaving are 1.5A and I_2 . So, $I_1 = 1.5 + I_2$ Putting $I_1 = 4A$, $4 = 1.5 + I_2 \Rightarrow I_2 = 4 - 1.5 = 2.5A$ For the third junction, I_2 enters. The currents leaving are 1.5A and I_3 . So, $I_2 = 1.5 + I_3$ Putting $I_2 = 2.5A$, $2.5 = 1.5 + I_3 \Rightarrow I_3 = 2.5 - 1.5 = 1A$

Benchmark Problem 2: KCET 2026 (Q#1)

Given below are two statements: Statement-I: The resistivity of a conductor is independent of its temperature. Statement-II: The resistivity of a semiconductor decreases with increase in temperature. Select the correct option.

- (A) Both Statement I and Statement II are false
- (B) Both Statement I and Statement II are true
- (C) Statement I is true but Statement II is false
- (D) Statement I is false but Statement II is true

Correct Answer: None

Solution Analysis: Statement-I is false. For a conductor, especially a metal, resistivity depends on temperature. When temperature increases, the ions in the metal vibrate more strongly. This makes it harder for free electrons to move, so the resistivity increases. So, resistivity of a conductor is not independent of temperature. Statement-II is true. For a semiconductor, when temperature increases, more electrons get enough energy to move to the conduction band. This increases the number of charge carriers, so conductivity increases. Hence, resistivity decreases with increase in temperature. So the correct option is (D).